

DLR-TomoSAR — Physics-constrained learning

five loss terms · curriculum · the unrolled solver

July 2026

Why constrain the solution space

- The labels are classical fits: the network's ceiling is the label quality — and Act IV showed the network happily leans on statistical shortcuts.
- Physics terms restrict predictions to profiles **consistent with the SAR signal model**:
 - today — as a regularizer: physical quantities recomputed from the predicted profile must match those of the GT profile;
 - next — as self-supervision against the *measured* covariance itself, removing the label ceiling entirely.
- Five terms implemented and theory-tested; every one differentiable end-to-end through the Gaussian-mixture curve reconstruction.

The shared forward operator

All terms use the vertical wavenumber and the steering matrix on the elevation grid:

$$k_z^{(i)} = \frac{4\pi b_i}{\lambda r_0 s}, \quad A_{in} = e^{j k_z^{(i)} \xi_n}$$

- Geometry resolved from campaign track parameters — not nominal defaults: per-pass perpendicular baselines, look angle, slant range.
- **Per-pixel k_z field** for range/azimuth-varying geometry, built at preprocessing and threaded dataset → trainer → loss.
- Steering sign settled empirically ($+k_z$: median peak error 0.67 m, within one Rayleigh cell).

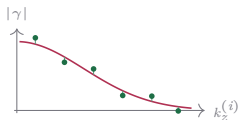
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n) e^{jk_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i |\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences,
baseline by baseline.

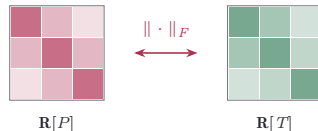
Cloude 2006 (PCT); same Fourier kernel as gamma-Net
(Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A} \text{diag}(P) \mathbf{A}^H \Delta \xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale,
entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros &
Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

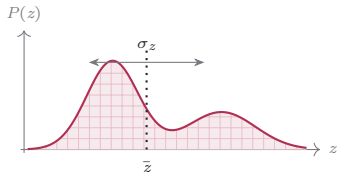
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n) \Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \quad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



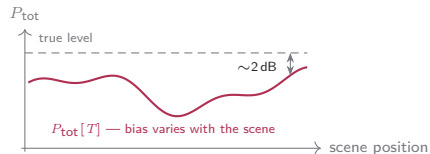
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\cdot] = \sum_n (\cdot)(\xi_n) \Delta\xi$$

$$\ell_{\text{pow}} = \left\langle (P_{\text{tot}}[P] - P_{\text{tot}}[T])^2 \right\rangle$$

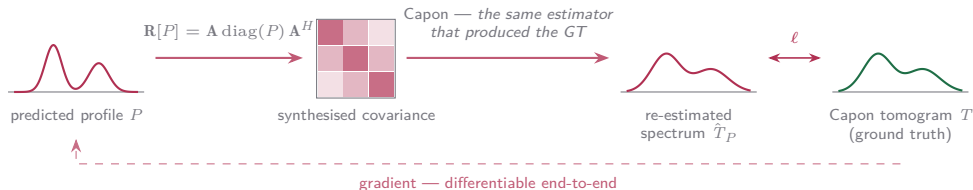


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = (\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n))^{-1}$$

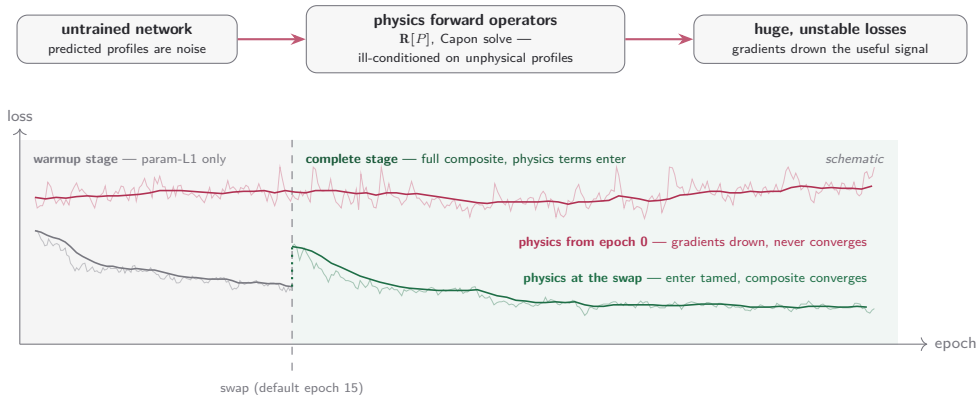


- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

Why the physics terms need a curriculum

The physics terms are the most sensitive consumers of the reconstructed profile — and at random initialisation they see only noise:



Swap mechanics, calibration — and status

Each detail is an anticipated failure mode:

- **Resets at the swap:** LR schedule origin, LR warmup, Adam moments, early-stopping counters — optimizer state calibrated to the old loss is stale under the new one.
- **Checkpoint-baseline reset:** best-loss tracking restarts across the non-comparable loss scales — otherwise the pre-swap best could never be beaten.
- **Stage inheritance:** the complete stage is the base configuration, warmup a derived pre-phase materialised at launch — the stages cannot drift.
- **Loss-scale probe:** every active term run at unit weight over probe batches, IQR-filtered, folded into the per-term weights — physics terms enter at calibrated magnitude, not guessed weights.

Status

Implemented, SAR-theory-verified, geometry wired end-to-end — the single-term $K=2$ sweep is in (next frames); the Capon-cycle arm, the most expensive of the five, is still queued.

The physics sweep at $K=2$ — every term rides low, the scale terms collapse high

objective	reconstruction		detection (matched)						
	curve $R^2 \uparrow$	cos med $\uparrow$	prec. $\uparrow$	recall $\uparrow$	F1 $\uparrow$	exact $\uparrow$	under $\downarrow$	over $\downarrow$	peak p95 $\downarrow$
param-L1 + cos	0.747 $\pm$.006	0.965 $\pm$.004	0.895 $\pm$.008	0.901 $\pm$.001	0.898 $\pm$.004	0.908 $\pm$.007	0.041 $\pm$.003	0.051 $\pm$.009	4.7 $\pm$ 0
+ coherence re-synthesis									
$w=0.01$	0.752 $\pm$.004	0.966 $\pm$.002	0.895 $\pm$.006	0.902 $\pm$.001	0.898 $\pm$.003	0.910 $\pm$.002	0.040 $\pm$.004	0.050 $\pm$.005	4.7 $\pm$ 0
$w=0.05$	0.752 $\pm$.006	0.966 $\pm$.001	0.893 $\pm$.005	0.902 $\pm$.001	0.898 $\pm$.002	0.911 $\pm$.003	0.039 $\pm$.003	0.051 $\pm$.004	4.7 $\pm$ 0
$w=0.25$	0.755 $\pm$.007	0.966 $\pm$.002	0.892 $\pm$.009	0.903 $\pm$.002	0.897 $\pm$.004	0.909 $\pm$.005	0.038 $\pm$.005	0.053 $\pm$.010	4.7 $\pm$ 0
$w=0.5$	0.748 $\pm$.009	0.965 $\pm$.003	0.886 $\pm$.013	0.903 $\pm$.001	0.895 $\pm$.007	0.911 $\pm$.010	0.032 $\pm$.006	0.057 $\pm$.014	4.7 $\pm$ 0
$w=1$	0.751 $\pm$.008	0.966 $\pm$.002	0.870 $\pm$.045	0.903 $\pm$.001	0.886 $\pm$.024	0.885 $\pm$.063	0.032 $\pm$.005	0.083 $\pm$.068	4.7 $\pm$ 0
+ covariance matching									
$w=0.01$	0.752 $\pm$.001	0.965 $\pm$.002	0.896 $\pm$.005	0.901 $\pm$.001	0.898 $\pm$.003	0.906 $\pm$.004	0.043 $\pm$.001	0.051 $\pm$.005	4.7 $\pm$ 0
$w=0.05$	0.748 $\pm$.001	0.966 $\pm$.002	0.899 $\pm$.005	0.902 $\pm$.001	0.900 $\pm$.002	0.909 $\pm$.004	0.043 $\pm$.002	0.048 $\pm$.005	4.7 $\pm$ 0
$w=0.25$	0.738 $\pm$.004	0.966 $\pm$.002	0.896 $\pm$.006	0.899 $\pm$.002	0.898 $\pm$.003	0.902 $\pm$.008	0.045 $\pm$.002	0.053 $\pm$.008	4.7 $\pm$ 0
$w=0.5$	0.727 $\pm$.011	0.965 $\pm$.002	0.900 $\pm$.001	0.898 $\pm$.002	0.899 $\pm$.001	0.907 $\pm$.003	0.046 $\pm$.003	0.047 $\pm$.001	4.7 $\pm$ 0
$w=1$	0.723 $\pm$.006	0.965 $\pm$.001	0.899 $\pm$.005	0.899 $\pm$.002	0.899 $\pm$.002	0.903 $\pm$.003	0.047 $\pm$.004	0.050 $\pm$.006	4.7 $\pm$ 0
+ moments									
$w=0.01$	0.753 $\pm$.003	0.966 $\pm$.002	0.895 $\pm$.006	0.902 $\pm$.001	0.898 $\pm$.003	0.910 $\pm$.004	0.040 $\pm$.005	0.050 $\pm$.006	4.7 $\pm$ 0
$w=0.05$	0.755 $\pm$.002	0.966 $\pm$.002	0.896 $\pm$.005	0.902 $\pm$.001	0.899 $\pm$.002	0.910 $\pm$.004	0.041 $\pm$.004	0.049 $\pm$.005	4.7 $\pm$ 0
$w=0.25$	0.751 $\pm$.002	0.965 $\pm$.002	0.897 $\pm$.004	0.901 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.003	0.044 $\pm$.000	0.049 $\pm$.004	4.7 $\pm$ 0
$w=0.5$	0.750 $\pm$.004	0.966 $\pm$.002	0.898 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.001	0.909 $\pm$.001	0.044 $\pm$.002	0.047 $\pm$.003	4.7 $\pm$ 0
$w=1$	0.746 $\pm$.005	0.966 $\pm$.001	0.897 $\pm$.003	0.901 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.003	0.043 $\pm$.001	0.049 $\pm$.003	4.7 $\pm$ 0
+ total power									
$w=0.01$	0.752 $\pm$.003	0.965 $\pm$.002	0.896 $\pm$.005	0.902 $\pm$.001	0.899 $\pm$.002	0.910 $\pm$.004	0.041 $\pm$.004	0.049 $\pm$.005	4.7 $\pm$ 0
$w=0.05$	0.751 $\pm$.005	0.966 $\pm$.001	0.897 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.002	0.044 $\pm$.001	0.048 $\pm$.002	4.7 $\pm$ 0
$w=0.25$	0.747 $\pm$.003	0.965 $\pm$.001	0.898 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.001	0.908 $\pm$.004	0.044 $\pm$.003	0.049 $\pm$.004	4.7 $\pm$ 0
$w=0.5$	0.737 $\pm$.015	0.965 $\pm$.002	0.900 $\pm$.005	0.899 $\pm$.002	0.899 $\pm$.003	0.905 $\pm$.006	0.048 $\pm$.004	0.047 $\pm$.006	4.7 $\pm$ 0
$w=1$	0.724 $\pm$.020	0.963 $\pm$.005	0.902 $\pm$.003	0.898 $\pm$.005	0.900 $\pm$.001	0.907 $\pm$.003	0.049 $\pm$.007	0.044 $\pm$.004	4.8 $\pm$ 3

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Row = loss (param-L1 + 0.5 cos + w phys) / (1.5 + w), one physics term per arm; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. Val losses do not compare across arms (different composites); curve MAE / PSNR / SSIM stay flat outside the collapse arms (next frames); peak p95 is pinned at 4.7. **bold** = best in column, marked only where best and worst differ by more than 5%. **Green**: coherence $w=0.25$ takes the best curve R^2 ; at $w=1$ its gate trades misses for false alarms (under 0.041 $\rightarrow$ 0.032, over 0.051 $\rightarrow$ 0.083; seed lottery: precision $\pm$.045, exact $\pm$.063). **Red** = absolute-scale collapse past $w=0.25$ (covariance, total power). cos med: median per-pixel cosine to GT: peak p95: 95th pct of the per-pixel peak error.

The $K=2$ physics detection by scatterer count — the set is untouched at every weight

objective	precision $\uparrow$			recall $\uparrow$			count acc pred $k \uparrow$	
	all	$k=1$	$k=2$	all	$k=1$	$k=2$	$k=1$	$k=2$
param-L1 + cos	0.895 $\pm$.008	0.935 $\pm$.010	0.828 $\pm$.006	0.901 $\pm$.001	0.999 $\pm$.000	0.763 $\pm$.002	0.944 $\pm$.004	0.814 $\pm$.024
+ <i>coherence re-synthesis</i>								
$w=0.01$	0.895 $\pm$.006	0.935 $\pm$.006	0.828 $\pm$.006	0.902 $\pm$.001	0.999 $\pm$.000	0.765 $\pm$.001	0.945 $\pm$.005	0.815 $\pm$.012
$w=0.05$	0.893 $\pm$.005	0.935 $\pm$.005	0.826 $\pm$.005	0.902 $\pm$.001	0.999 $\pm$.000	0.765 $\pm$.002	0.947 $\pm$.004	0.814 $\pm$.011
$w=0.25$	0.892 $\pm$.009	0.932 $\pm$.011	0.826 $\pm$.006	0.903 $\pm$.002	0.999 $\pm$.000	0.766 $\pm$.004	0.948 $\pm$.006	0.808 $\pm$.023
$w=0.5$	0.886 $\pm$.013	0.928 $\pm$.016	0.818 $\pm$.010	0.903 $\pm$.001	0.999 $\pm$.000	0.767 $\pm$.003	0.955 $\pm$.008	0.803 $\pm$.033
$w=1$	0.870 $\pm$.045	0.902 $\pm$.067	0.818 $\pm$.009	0.903 $\pm$.001	0.999 $\pm$.000	0.768 $\pm$.001	0.954 $\pm$.002	0.755 $\pm$.122
+ <i>covariance matching</i>								
$w=0.01$	0.896 $\pm$.005	0.935 $\pm$.006	0.831 $\pm$.004	0.901 $\pm$.001	0.999 $\pm$.000	0.761 $\pm$.002	0.943 $\pm$.003	0.812 $\pm$.016
$w=0.05$	0.899 $\pm$.005	0.938 $\pm$.006	0.834 $\pm$.003	0.902 $\pm$.001	0.999 $\pm$.000	0.763 $\pm$.003	0.943 $\pm$.002	0.819 $\pm$.015
$w=0.25$	0.896 $\pm$.006	0.932 $\pm$.009	0.835 $\pm$.002	0.899 $\pm$.002	0.999 $\pm$.000	0.758 $\pm$.005	0.944 $\pm$.003	0.803 $\pm$.023
$w=0.5$	0.900 $\pm$.001	0.939 $\pm$.002	0.836 $\pm$.002	0.898 $\pm$.002	0.999 $\pm$.000	0.755 $\pm$.005	0.943 $\pm$.005	0.820 $\pm$.005
$w=1$	0.899 $\pm$.005	0.936 $\pm$.007	0.837 $\pm$.002	0.899 $\pm$.002	0.999 $\pm$.000	0.756 $\pm$.005	0.942 $\pm$.007	0.812 $\pm$.015
+ <i>moments</i>								
$w=0.01$	0.895 $\pm$.006	0.936 $\pm$.007	0.827 $\pm$.007	0.902 $\pm$.001	0.999 $\pm$.000	0.764 $\pm$.002	0.945 $\pm$.006	0.816 $\pm$.015
$w=0.05$	0.896 $\pm$.005	0.937 $\pm$.006	0.828 $\pm$.006	0.902 $\pm$.001	0.999 $\pm$.000	0.764 $\pm$.003	0.944 $\pm$.005	0.817 $\pm$.015
$w=0.25$	0.897 $\pm$.004	0.937 $\pm$.004	0.831 $\pm$.004	0.901 $\pm$.001	0.999 $\pm$.000	0.761 $\pm$.003	0.941 $\pm$.001	0.817 $\pm$.011
$w=0.5$	0.898 $\pm$.003	0.939 $\pm$.004	0.832 $\pm$.003	0.900 $\pm$.001	0.999 $\pm$.000	0.761 $\pm$.003	0.942 $\pm$.002	0.821 $\pm$.009
$w=1$	0.897 $\pm$.003	0.937 $\pm$.004	0.832 $\pm$.003	0.901 $\pm$.001	0.999 $\pm$.000	0.761 $\pm$.002	0.943 $\pm$.001	0.817 $\pm$.009
+ <i>total power</i>								
$w=0.01$	0.896 $\pm$.005	0.937 $\pm$.006	0.828 $\pm$.006	0.902 $\pm$.001	0.999 $\pm$.000	0.763 $\pm$.001	0.944 $\pm$.005	0.818 $\pm$.013
$w=0.05$	0.897 $\pm$.003	0.938 $\pm$.003	0.831 $\pm$.004	0.900 $\pm$.001	0.999 $\pm$.000	0.761 $\pm$.003	0.941 $\pm$.001	0.819 $\pm$.007
$w=0.25$	0.898 $\pm$.003	0.937 $\pm$.004	0.833 $\pm$.004	0.900 $\pm$.001	0.999 $\pm$.000	0.761 $\pm$.003	0.942 $\pm$.002	0.817 $\pm$.011
$w=0.5$	0.900 $\pm$.005	0.939 $\pm$.007	0.834 $\pm$.003	0.899 $\pm$.002	0.999 $\pm$.000	0.757 $\pm$.006	0.937 $\pm$.006	0.820 $\pm$.018
$w=1$	0.902 $\pm$.003	0.943 $\pm$.005	0.834 $\pm$.002	0.898 $\pm$.005	0.999 $\pm$.000	0.755 $\pm$.011	0.934 $\pm$.008	0.829 $\pm$.009

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Row = one physics term at weight w ; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. All rows matched (Hungarian pairing); k = true scatterer count of the pixel. **bold** = best in column, marked only where best and worst differ by more than 5% — almost nothing clears it at any weight: the physics terms shape the profile, not the scatterer set. The exception is the high-weight coherence gate: $k=2$ recall creeps up (0.763 $\rightarrow$ 0.768) while the claimed pairs destabilise ($k=1$ precision seed lottery $\pm$.067, pred $k=2$ accuracy $\pm$.122). Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k .

The $K=2$ physics component errors — covariance buys position, coherence pays it

objective	μ MAE $\downarrow$			σ MAE $\downarrow$			a MAE $\downarrow$		
	all	$k=1$	$k=2$	all	$k=1$	$k=2$	all	$k=1$	$k=2$
param-L1 + cos	1.71 $\pm$.07	0.36 $\pm$.01	3.78 $\pm$.14	0.774 $\pm$.011	0.193 $\pm$.002	1.668 $\pm$.023	0.182 $\pm$.004	0.058 $\pm$.003	0.373 $\pm$.006
+ coherence re-synthesis									
$w=0.01$	1.70 $\pm$.06	0.35 $\pm$.01	3.78 $\pm$.12	0.770 $\pm$.009	0.194 $\pm$.003	1.654 $\pm$.024	0.180 $\pm$.003	0.057 $\pm$.002	0.369 $\pm$.004
$w=0.05$	1.72 $\pm$.05	0.35 $\pm$.01	3.82 $\pm$.11	0.779 $\pm$.014	0.195 $\pm$.002	1.673 $\pm$.035	0.180 $\pm$.003	0.057 $\pm$.003	0.369 $\pm$.004
$w=0.25$	1.72 $\pm$.06	0.35 $\pm$.00	3.82 $\pm$.13	0.778 $\pm$.011	0.192 $\pm$.003	1.675 $\pm$.016	0.182 $\pm$.005	0.058 $\pm$.004	0.373 $\pm$.005
$w=0.5$	1.80 $\pm$.10	0.35 $\pm$.01	4.00 $\pm$.22	0.804 $\pm$.017	0.191 $\pm$.002	1.730 $\pm$.033	0.183 $\pm$.004	0.058 $\pm$.003	0.372 $\pm$.004
$w=1$	1.81 $\pm$.12	0.35 $\pm$.00	4.02 $\pm$.26	0.809 $\pm$.020	0.194 $\pm$.006	1.737 $\pm$.035	0.185 $\pm$.006	0.061 $\pm$.008	0.373 $\pm$.004
+ covariance matching									
$w=0.01$	1.66 $\pm$.04	0.35 $\pm$.01	3.70 $\pm$.09	0.755 $\pm$.014	0.192 $\pm$.004	1.626 $\pm$.032	0.182 $\pm$.003	0.058 $\pm$.003	0.374 $\pm$.004
$w=0.05$	1.65 $\pm$.02	0.35 $\pm$.01	3.66 $\pm$.05	0.754 $\pm$.007	0.194 $\pm$.002	1.620 $\pm$.017	0.187 $\pm$.005	0.061 $\pm$.004	0.381 $\pm$.007
$w=0.25$	1.62 $\pm$.01	0.34 $\pm$.01	3.62 $\pm$.04	0.725 $\pm$.011	0.186 $\pm$.002	1.566 $\pm$.024	0.196 $\pm$.004	0.065 $\pm$.004	0.400 $\pm$.006
$w=0.5$	1.61 $\pm$.01	0.34 $\pm$.01	3.61 $\pm$.03	0.724 $\pm$.014	0.186 $\pm$.003	1.567 $\pm$.035	0.205 $\pm$.005	0.069 $\pm$.004	0.418 $\pm$.010
$w=1$	1.61 $\pm$.02	0.35 $\pm$.01	3.61 $\pm$.04	0.728 $\pm$.018	0.189 $\pm$.008	1.575 $\pm$.039	0.214 $\pm$.012	0.071 $\pm$.005	0.439 $\pm$.024
+ moments									
$w=0.01$	1.71 $\pm$.06	0.35 $\pm$.01	3.79 $\pm$.14	0.772 $\pm$.017	0.195 $\pm$.003	1.657 $\pm$.041	0.180 $\pm$.003	0.057 $\pm$.002	0.369 $\pm$.004
$w=0.05$	1.69 $\pm$.04	0.35 $\pm$.01	3.75 $\pm$.09	0.771 $\pm$.018	0.195 $\pm$.003	1.657 $\pm$.043	0.182 $\pm$.003	0.058 $\pm$.003	0.372 $\pm$.002
$w=0.25$	1.68 $\pm$.04	0.35 $\pm$.01	3.72 $\pm$.09	0.763 $\pm$.005	0.195 $\pm$.003	1.643 $\pm$.015	0.186 $\pm$.004	0.060 $\pm$.003	0.380 $\pm$.005
$w=0.5$	1.66 $\pm$.03	0.35 $\pm$.01	3.69 $\pm$.07	0.759 $\pm$.007	0.194 $\pm$.002	1.634 $\pm$.016	0.190 $\pm$.004	0.062 $\pm$.004	0.387 $\pm$.005
$w=1$	1.66 $\pm$.02	0.35 $\pm$.01	3.69 $\pm$.04	0.760 $\pm$.008	0.196 $\pm$.004	1.633 $\pm$.012	0.192 $\pm$.002	0.063 $\pm$.003	0.391 $\pm$.003
+ total power									
$w=0.01$	1.70 $\pm$.05	0.35 $\pm$.01	3.77 $\pm$.10	0.771 $\pm$.019	0.196 $\pm$.004	1.656 $\pm$.045	0.180 $\pm$.003	0.057 $\pm$.003	0.370 $\pm$.003
$w=0.05$	1.67 $\pm$.03	0.35 $\pm$.00	3.72 $\pm$.08	0.761 $\pm$.005	0.194 $\pm$.003	1.639 $\pm$.015	0.184 $\pm$.004	0.059 $\pm$.004	0.377 $\pm$.004
$w=0.25$	1.65 $\pm$.03	0.35 $\pm$.01	3.66 $\pm$.05	0.753 $\pm$.011	0.195 $\pm$.004	1.619 $\pm$.016	0.188 $\pm$.005	0.060 $\pm$.005	0.385 $\pm$.006
$w=0.5$	1.63 $\pm$.02	0.35 $\pm$.01	3.62 $\pm$.04	0.762 $\pm$.022	0.197 $\pm$.006	1.646 $\pm$.052	0.192 $\pm$.004	0.062 $\pm$.003	0.396 $\pm$.007
$w=1$	1.63 $\pm$.02	0.35 $\pm$.01	3.63 $\pm$.03	0.777 $\pm$.021	0.205 $\pm$.008	1.672 $\pm$.055	0.198 $\pm$.005	0.063 $\pm$.003	0.410 $\pm$.012

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Row = one physics term at weight w ; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. Component errors over matched pairs only (Hungarian pairing); k = true scatterer count of the pixel. **bold** = best in column, marked only where best and worst differ by more than 5%. **Covariance at $w=0.05$ takes the best μ** (1.65 vs 1.71); **full-weight coherence pays position and width** (μ 1.81, σ 0.809) — the wider gate matches harder pairs; amplitude never moves. μ , σ in elevation units; a in linear amplitude.

The $K=2$ physics curve fidelity — flat everywhere except the collapse arms

objective	curve			SSIM vs GT			peak
	MAE ↓	RMSE ↓	PSNR [dB] ↑	elev ↑	range ↑	azim ↑	mean ↓
param-L1 + cos	0.0370 ±.0005	0.217 ±.002	52.63 ±.10	0.9671 ±.0004	0.9510 ±.0008	0.9610 ±.0006	1.23 ±.02
+ <i>coherence re-synthesis</i>							
$w=0.01$	0.0367 ±.0003	0.215 ±.002	52.71 ±.07	0.9673 ±.0002	0.9513 ±.0006	0.9613 ±.0004	1.22 ±.01
$w=0.05$	0.0368 ±.0003	0.215 ±.003	52.72 ±.10	0.9673 ±.0002	0.9512 ±.0005	0.9612 ±.0004	1.22 ±.01
$w=0.25$	0.0366 ±.0003	0.213 ±.003	52.78 ±.13	0.9674 ±.0002	0.9516 ±.0004	0.9615 ±.0003	1.22 ±.01
$w=0.5$	0.0370 ±.0006	0.217 ±.004	52.65 ±.16	0.9670 ±.0006	0.9504 ±.0015	0.9607 ±.0010	1.23 ±.02
$w=1$	0.0371 ±.0007	0.215 ±.003	52.70 ±.14	0.9670 ±.0006	0.9499 ±.0027	0.9606 ±.0014	1.22 ±.01
+ <i>covariance matching</i>							
$w=0.01$	0.0366 ±.0001	0.215 ±.001	52.72 ±.02	0.9675 ±.0002	0.9520 ±.0004	0.9615 ±.0003	1.22 ±.01
$w=0.05$	0.0365 ±.0002	0.217 ±.000	52.65 ±.02	0.9675 ±.0002	0.9521 ±.0004	0.9616 ±.0002	1.21 ±.01
$w=0.25$	0.0368 ±.0002	0.221 ±.002	52.49 ±.06	0.9673 ±.0001	0.9516 ±.0003	0.9612 ±.0002	1.23 ±.01
$w=0.5$	0.0375 ±.0007	0.225 ±.005	52.31 ±.18	0.9669 ±.0005	0.9509 ±.0008	0.9606 ±.0007	1.23 ±.01
$w=1$	0.0377 ±.0005	0.227 ±.002	52.24 ±.09	0.9670 ±.0003	0.9510 ±.0006	0.9605 ±.0005	1.24 ±.01
+ <i>moments</i>							
$w=0.01$	0.0366 ±.0002	0.215 ±.001	52.73 ±.05	0.9674 ±.0002	0.9516 ±.0006	0.9614 ±.0004	1.22 ±.01
$w=0.05$	0.0365 ±.0002	0.214 ±.001	52.76 ±.04	0.9674 ±.0002	0.9519 ±.0004	0.9615 ±.0003	1.22 ±.01
$w=0.25$	0.0366 ±.0002	0.215 ±.001	52.71 ±.04	0.9674 ±.0002	0.9519 ±.0004	0.9615 ±.0004	1.22 ±.01
$w=0.5$	0.0365 ±.0003	0.216 ±.002	52.69 ±.07	0.9675 ±.0002	0.9520 ±.0005	0.9616 ±.0004	1.22 ±.01
$w=1$	0.0367 ±.0003	0.218 ±.002	52.61 ±.08	0.9675 ±.0003	0.9519 ±.0004	0.9614 ±.0003	1.22 ±.01
+ <i>total power</i>							
$w=0.01$	0.0367 ±.0002	0.215 ±.001	52.71 ±.06	0.9673 ±.0002	0.9517 ±.0005	0.9614 ±.0004	1.22 ±.00
$w=0.05$	0.0366 ±.0001	0.215 ±.002	52.70 ±.08	0.9674 ±.0001	0.9519 ±.0004	0.9615 ±.0003	1.22 ±.01
$w=0.25$	0.0367 ±.0003	0.217 ±.001	52.64 ±.05	0.9674 ±.0002	0.9519 ±.0005	0.9614 ±.0004	1.23 ±.01
$w=0.5$	0.0373 ±.0011	0.221 ±.006	52.48 ±.24	0.9669 ±.0009	0.9512 ±.0013	0.9608 ±.0011	1.22 ±.01
$w=1$	0.0377 ±.0015	0.227 ±.008	52.27 ±.31	0.9667 ±.0011	0.9508 ±.0016	0.9605 ±.0014	1.23 ±.04

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. Row = one physics term at weight w ; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px. **bold** = best in column, marked only where best and worst differ by more than 5% — no column clears it: curve fidelity and structure are flat at every gentle weight, and only degrade on **the absolute-scale collapse arms** (covariance / total power at $w \geq 0.5$: RMSE 0.217 → 0.227, SSIM range 0.951 → 0.944). Peak location is pinned everywhere: median 0.67 and p95 4.7 in every arm; the mean moves in the second decimal. peak in elevation units.

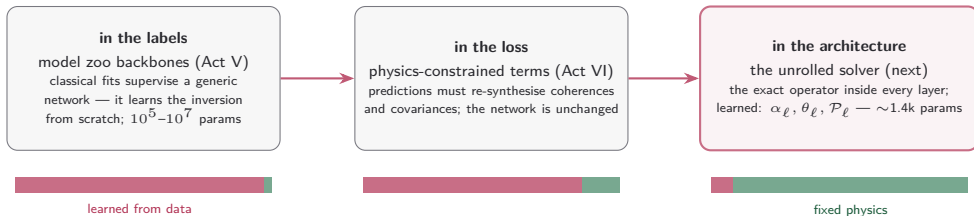
The $K=2$ physics per-pixel tail — descriptor and covariance terms cap the blow-ups

objective	per-pixel $R^2 \uparrow$					per-pixel cosine $\uparrow$		
	med	mean	p_5	p_1	min	med	mean	p_1
param-L1 + cos	0.888 $\pm$.007	-1.2 $\pm$.8	+0.3 $\pm$.1	-14.9 $\pm$ 11.4	-13.2k $\pm$ 3.3k	0.965 $\pm$.004	0.934 $\pm$.003	0.588 $\pm$.011
+ coherence re-synthesis								
$w=0.01$	0.889 $\pm$.005	-1.0 $\pm$.8	+0.3 $\pm$.1	-13.5 $\pm$ 9.2	-12.7k $\pm$ 7.4k	0.966 $\pm$.002	0.935 $\pm$.002	0.590 $\pm$.006
$w=0.05$	0.888 $\pm$.004	-1.2 $\pm$.9	+0.3 $\pm$.1	-14.8 $\pm$ 8.1	-11.9k $\pm$ 8.0k	0.966 $\pm$.001	0.935 $\pm$.002	0.587 $\pm$.008
$w=0.25$	0.891 $\pm$.006	-0.9 $\pm$.5	+0.3 $\pm$.0	-12.4 $\pm$ 4.0	-12.2k $\pm$ 4.8k	0.966 $\pm$.002	0.936 $\pm$.001	0.599 $\pm$.009
$w=0.5$	0.888 $\pm$.010	-9.6 $\pm$ 12.0	+0.2 $\pm$.2	-70.2 $\pm$ 80.2	-66.3k $\pm$ 77.4k	0.965 $\pm$.003	0.935 $\pm$.002	0.598 $\pm$.006
$w=1$	0.870 $\pm$.049	-3.7 $\pm$ 4.3	-0.9 $\pm$ 2.6	-50.2 $\pm$ 71.0	-22.5k $\pm$ 11.5k	0.966 $\pm$.002	0.936 $\pm$.001	0.601 $\pm$.007
+ covariance matching								
$w=0.01$	0.889 $\pm$.008	+0.7 $\pm$.0	+0.4 $\pm$.0	-0.7 $\pm$.2	-4.3k $\pm$ 0.2k	0.965 $\pm$.002	0.934 $\pm$.001	0.575 $\pm$.013
$w=0.05$	0.889 $\pm$.009	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.5 $\pm$.1	-4.2k $\pm$ 0.2k	0.966 $\pm$.002	0.934 $\pm$.001	0.574 $\pm$.007
$w=0.25$	0.884 $\pm$.006	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.4 $\pm$.0	-4.6k $\pm$ 0.1k	0.966 $\pm$.002	0.932 $\pm$.001	0.544 $\pm$.014
$w=0.5$	0.880 $\pm$.007	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.4 $\pm$.0	-4.6k $\pm$ 0.0k	0.965 $\pm$.002	0.931 $\pm$.001	0.529 $\pm$.011
$w=1$	0.882 $\pm$.004	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.3 $\pm$.0	-4.6k $\pm$ 0.1k	0.965 $\pm$.001	0.929 $\pm$.001	0.510 $\pm$.025
+ moments								
$w=0.01$	0.888 $\pm$.008	-0.1 $\pm$.5	+0.3 $\pm$.1	-7.1 $\pm$ 6.6	-6.2k $\pm$ 2.2k	0.966 $\pm$.002	0.935 $\pm$.001	0.588 $\pm$.004
$w=0.05$	0.888 $\pm$.009	+0.5 $\pm$.2	+0.4 $\pm$.0	-2.2 $\pm$ 1.1	-4.4k $\pm$ 1.2k	0.966 $\pm$.002	0.935 $\pm$.001	0.584 $\pm$.006
$w=0.25$	0.888 $\pm$.008	+0.7 $\pm$.0	+0.4 $\pm$.0	-0.8 $\pm$.2	-4.2k $\pm$ 0.1k	0.965 $\pm$.002	0.934 $\pm$.001	0.578 $\pm$.004
$w=0.5$	0.887 $\pm$.007	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.5 $\pm$.0	-4.3k $\pm$ 0.3k	0.966 $\pm$.002	0.934 $\pm$.001	0.570 $\pm$.009
$w=1$	0.886 $\pm$.006	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.4 $\pm$.0	-4.4k $\pm$ 0.3k	0.966 $\pm$.001	0.933 $\pm$.001	0.562 $\pm$.008
+ total power								
$w=0.01$	0.889 $\pm$.009	+0.5 $\pm$.2	+0.3 $\pm$.0	-2.6 $\pm$ 1.2	-3.6k $\pm$ 0.2k	0.965 $\pm$.002	0.935 $\pm$.002	0.584 $\pm$.005
$w=0.05$	0.889 $\pm$.007	+0.7 $\pm$.0	+0.4 $\pm$.0	-0.9 $\pm$.4	-4.1k $\pm$ 0.1k	0.966 $\pm$.001	0.934 $\pm$.001	0.579 $\pm$.008
$w=0.25$	0.888 $\pm$.007	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.5 $\pm$.3	-4.3k $\pm$ 0.3k	0.965 $\pm$.001	0.933 $\pm$.001	0.563 $\pm$.016
$w=0.5$	0.887 $\pm$.009	+0.8 $\pm$.0	+0.4 $\pm$.0	-0.5 $\pm$.3	-4.2k $\pm$ 0.3k	0.965 $\pm$.002	0.931 $\pm$.002	0.551 $\pm$.010
$w=1$	0.885 $\pm$.011	+0.7 $\pm$.0	+0.4 $\pm$.0	-0.8 $\pm$.5	-4.2k $\pm$ 0.2k	0.963 $\pm$.005	0.929 $\pm$.005	0.529 $\pm$.026

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Row = one physics term at weight w ; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. pixel R^2 min and its spread in thousands (k). **bold** = best in column, marked only where best and worst differ by more than 5%. The typical pixel never moves (median and cosine flat); the tail is the story: catastrophic pixels wreck the no-physics mean (-1.17 , $p_1 -14.9$, min $-13.2k$); **covariance caps the blow-ups at every weight, improving as w grows** (mean $+0.8$ at $w=1$); moments and total power insure at low weight (best min $-3.6k$). **Coherence is NOT tail insurance**: worse than the baseline at $w>0.5$ (mean -9.6 , min $-66k$) — it shapes coherences, not pixel-domain outliers.

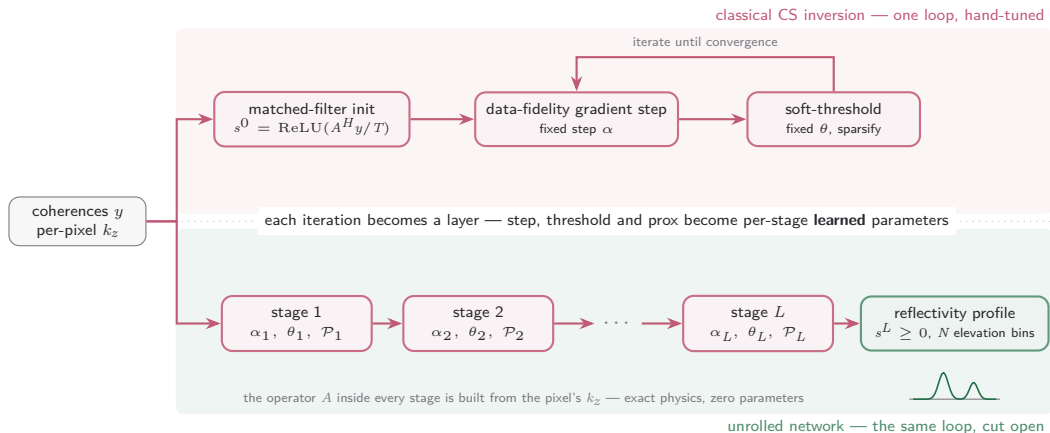
Context: where does the physics live?

Three places the SAR signal model can enter a learned inversion — the project now covers all three:



The unrolled family probes the far end of the axis — **the question it answers:** how much of the zoo's capacity is spent re-learning physics we already know?

The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \text{Re}[A^H(y - As^\ell)] / L_{\text{lip}}, \quad s^{\ell+1} = \text{ReLU}(\mathcal{P}_\ell(r^\ell) - \theta_\ell)$$

γ -Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\text{lip}} = TN \text{d}z^2$ normalises the step. Default $L=8$: $\sim 1.4\text{k}$ parameters.

Adapting γ -Net — a different unknown, a per-pixel operator

γ -Net — one scene geometry, weights learned



a fixed matrix can be learned — an operator that changes at every pixel cannot



this project — per-pixel geometry, physics exact, scalars learned

γ -Net inverts $g = \mathbf{R}\gamma + \varepsilon, \quad \gamma \in \mathbb{C}^N$

this project inverts $y = A(x)s, \quad s \in \mathbb{R}_{\geq 0}^N$

g : one complex look per track (first-order); y : per-track coherences (second-order) — averaging interferograms cancels the speckle phase, so the recoverable object is the power profile s , not the complex reflectivity γ . ε : measurement noise; T : tracks; N : elevation bins; $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation.

Adapting γ -Net — mechanism by mechanism

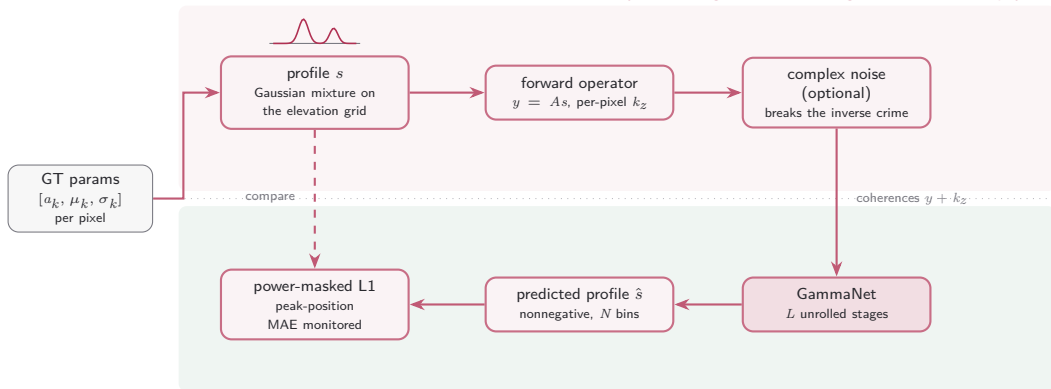
mechanism	γ -Net (Qian et al. 2022)	this project	why
iteration	gradient step $\rightarrow$ shrinkage, L stages	= kept	the unrolling premise itself
supervision	train on simulated measurements	= kept	the exact operator generates the pairs
unknown	complex reflectivity $\gamma \in \mathbb{C}^N$	$\rightarrow$ power profile $s \in \mathbb{R}_{\geq 0}^N$	speckle phase averages out of coherences
measurements	single-look stack g	$\rightarrow$ per-track coherences y	the pipeline is second-order end to end
operator	one fixed $\mathbf{R}$, scene-wide	$\rightarrow$ $A(x)$ from the pixel's k_z	airborne geometry varies across the scene
learned weights	$\mathbf{W}^\ell$ per stage, coupled to $\mathbf{R}$	$\rightarrow$ none — scalars α_ℓ, θ_ℓ	a per-pixel matrix cannot be learned
shrinkage	support selection + piecewise-linear function	$\rightarrow$ prox $\mathcal{P}_\ell$, then $\text{ReLU}(\cdot - \theta_\ell)$	the prior must respect $s \geq 0$
model order	Bayesian information criterion after inversion	$\rightarrow$ downstream parameter extraction	the pipeline already fits K scatterers

The contract every change preserves: **the physics stays exact inside the network — only the prior is learned.**

support selection: the largest-magnitude entries bypass the threshold at each stage; piecewise-linear function: γ -Net's learned replacement for soft-thresholding; coupled weights: $\mathbf{W}_2^\ell = \mathbf{I} - \mathbf{W}_1^\ell \mathbf{R}$ halves the free parameters per stage. γ -Net reaches its accuracy at 12 stages; default here $L = 8$. **green** = exact physics carried by the operator · **red** = learned · grey = kept from γ -Net.

Training the unrolled model against the exact operator

synthesis — ground truth through the exact forward physics



inversion — the unrolled network runs it backwards

Phase 1 (implemented): learn to invert the *exact* operator on synthesised coherences. Phase 2 (pending): measured covariances from the stack — no label ceiling.

Isolated family — own registry, trainer and entry point (`train_unrolled`); optimizer splits steps/thresholds from the prox; no AMP, the operator is complex-valued.

Why it exists: the physics-native comparison point — if $\sim 1.4k$ parameters of encoded physics match capacity-matched vision backbones, those backbones are mostly re-learning known physics.